Calculus (1) Quiz 1

Feb. 23 2026

考試時間: 50 min　　本試卷總分: 100 分　　實得分數

系所／班級: 運動系　　姓名: 張翼德　　學號: 415900797

Note: No mobile devices. Write inside the box only; no credit for answers outside. L'Hôpital's rule is prohibited.

1. (20 分) 計算以下極極限。

(1) (10 分) $\lim_{x\to 2} 3x^2 - x + 1$.

[Q1-1]

俺不會

(2) (10 分) $\lim_{x\to 2} \frac{x^2-4}{x-2}$

[Q1-2]

俺不會

2. (20 分) 找出常數 c 的值，使得以下函數在 $x = 3$ 處連續: $f(x) = \begin{cases} cx^2 - 1 & \text{if } x \le 3 \\ 2x + 5 & \text{if } x > 3 \end{cases}$

[Q2]

俺不會

3. (20 分) 計算以下極限: $\lim_{x\to 0} \frac{\sin(5x)}{x}$

[Q3]

俺不會

4. **(20 分)** 計算以下極限值: $\lim_{x\to 0} \frac{1-\cos(x^2)}{x^3 \sin(x)}$

[Q4]

俺不會

5. **(20 分)** Prove $\lim_{x\to 2} x^2 = 4$ by definition.

[Q5]

俺不會

Calculus (1) Quiz 1

Feb. 23 2026

考試時間: 50 min　　本試卷總分: 100 分　　實得分數

系所／班級: 數學一　　姓名: 馬春花　　學號: 415900579

Note: No mobile devices. Write inside the box only; no credit for answers outside. L'Hôpital's rule is prohibited.

$\frac{f(x)-f(2)}{x-2}$

$\frac{f(a+h)-f(a)}{h}$

1. (20 分) 計算以下極極限。

(1) (10 分) $\lim_{x\to 2} 3x^2 - x + 1$.

$f(x)$

[Q1-1]

$3\cdot 2^2 - 2 + 1 = 11$

$3x^2 - x - 10$

$\lim_{x\to 2}\frac{f(x)-f(2)}{x-2} = \lim_{x\to 2}\frac{(3x^2-x+1)-11}{x-2}$

$= \lim_{x\to 2}\frac{(x-2)(3x+5)}{x-2}$

$= 3\cdot 2+5$

$= 11$ ※

(2) (10 分) $\lim_{x\to 2}\frac{x^2-4}{x-2}$

[Q1-2]

$\lim_{x\to 2}\frac{(x+2)(x-2)}{x-2}$

$= 2+2$

$= 4$ ※

2. (20 分) 找出常數 c 的值，使得以下函數在 $x=3$ 處連續: $f(x) = \begin{cases} cx^2 - 1 & \text{if } x \le 3 \\ 2x+5 & \text{if } x > 3 \end{cases}$

[Q2]

$\lim_{x\to 3^+}(2x+5) = 11$

$\lim_{x\to 3^-}(cx^2-1) = 11$

$= 9c-1$

$\therefore c = \frac{12}{9} = \frac{4}{3}$ ※

3. (20 分) 計算以下極限: $\lim_{x\to 0}\frac{\sin(5x)}{x}$

$\lim_{x\to 0} \frac{\sin(5x)}{x}$

[Q3]

$$\lim_{x\to 0} \frac{\sin(5x)}{x} = 5 \cdot \lim_{x\to 0} \frac{\sin(5x)}{5x}$$
$$= 5 \cdot 1$$
$$= 5$$

4. (20 分) 計算以下極限值: $\lim_{x\to 0} \frac{1-\cos(x^2)}{x^3 \sin(x)}$ $\lim_{x\to 0} \frac{1-\cos t}{t^2} = \frac{1}{2}$

[Q4]

$$\lim_{x\to 0} \frac{(1-\cos x^2)\cdot x}{x^4 \cdot \sin x}$$
$$= \lim_{x\to 0} \frac{1-\cos x^2}{x^4} \cdot \lim_{x\to 0} \frac{x}{\sin x}$$
$$= \frac{1}{2} \cdot 1$$
$$= \frac{1}{2}$$

5. (20 分) Prove $\lim_{x\to 2} x^2 = 4$ by definition.

[Q5]

$f(x) = x^2$

$$\lim_{x\to 2} \frac{f(x)-f(2)}{x-2} = \lim_{x\to 2} \frac{x^2-4}{x-2} = \lim_{x\to 2} (x+2) = 4$$

$\square$

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考試時間: 50 min　　本試卷總分: 100 分　　實得分數

系所／班級: 公行系　　姓名: 胡一刀　　學號: 415100010

Note: No mobile devices. Write inside the box only; no credit for answers outside. L'Hôpital's rule is prohibited.

1. (20 分) 計算以下極極限。

(1) (10 分) $\lim_{x\to 2} 3x^2 - x + 1$.

[Q1-1]

$\lim_{x\to 2} 3x^2 - x + 1$

$= 3\cdot 2^2 - 2 + 1$

$= 12 - 2 + 1$

$= 11$

(2) (10 分) $\lim_{x\to 2} \frac{x^2-4}{x-2}$

[Q1-2]

$\lim_{x\to 2} \frac{(x-2)(x+2)}{x-2}$

$= \lim_{x\to 2} (x+2)$

$= 2+2 = 4$

2. (20 分) 找出常數 c 的值，使得以下函數在 $x = 3$ 處連續: $f(x) = \begin{cases} cx^2 - 1 & \text{if } x \le 3 \\ 2x + 5 & \text{if } x > 3 \end{cases}$

[Q2]

$x \le 3.\ f(3) = 9c - 1$

$x > 3.\ f(3^+) = 6 + 5 = 11$

$9c - 1 = 11$

$9c = 12$

$c = \frac{12}{9} = \frac{4}{3}$

3. (20 分) 計算以下極限: $\lim_{x\to 0} \frac{\sin(5x)}{x}$

[Q3]

$$\lim_{x\to 0} \frac{\sin 5x}{x} = \lim_{x\to 0} \frac{\sin 5x}{5x} \cdot 5$$

$$= 1 \cdot 5 = 5$$

4. (20 分) 計算以下極限值: $\lim_{x\to 0} \frac{1-\cos(x^2)}{x^3 \sin(x)}$

[Q4]

$$\lim_{x\to 0} \frac{1-\cos(x^2)}{x^3 \sin x}$$

$$= \lim_{x\to 0} \frac{\sin^2 x}{x^3 \sin x} = \infty$$

5. (20 分) Prove $\lim_{x\to 2} x^2 = 4$ by definition.

[Q5]

Given $\varepsilon > 0$. choose $\delta = \min\{\frac{\varepsilon}{5}, 1\}$

s.t. $0 < |x-2| < \delta$

$\Rightarrow |x^2-4| = |x-2||x+2|$

$< 5\delta = \varepsilon$ ★

if $\delta \le 1$.

$|x-2| < 1$

$-1 < x-2 < 1$

$3 < x+2 < 5$

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Feb. 23 2026

考試時間: 50 min　　本試卷總分: 100 分　　實得分數

系所／班級: 管科系　　姓名: 周伯通　　學號: 414201665

Note: No mobile devices. Write inside the box only; no credit for answers outside. L'Hôpital's rule is prohibited.

1. (20 分) 計算以下極極限。

(1) (10 分) $\lim_{x\to 2} 3x^2 - x + 1$.

[Q1-1]

$\lim_{x\to 2} 3x^2 - x + 1$

$= 3\cdot 4 - 2 + 1 = 11$

(2) (10 分) $\lim_{x\to 2} \frac{x^2-4}{x-2}$

[Q1-2]

$\lim_{x\to 2} \frac{x^2-4}{x-2} = \frac{0}{0} = 1$

2. (20 分) 找出常數 c 的值，使得以下函數在 $x = 3$ 處連續: $f(x) = \begin{cases} cx^2 - 1 & \text{if } x \le 3 \\ 2x + 5 & \text{if } x > 3 \end{cases}$

[Q2]

$f(3) = \lim_{x\to 3^-} cx^2 - 1 = c\cdot 9 - 1$

$= 6 + 5$

$c = \frac{12}{9}$

3. (20 分) 計算以下極限: $\lim_{x\to 0} \frac{\sin(5x)}{x}$

[Q3]

$$\lim_{x\to 0} \frac{\sin 5x}{x} = 1 \quad \because \lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1$$

4. (20 分) 計算以下極限值: $\lim_{x\to 0} \frac{1-\cos(x^2)}{x^3\sin(x)}$

[Q4]

$$\lim_{x\to 0} \frac{1-\cos(x^2)}{x^3 \sin x} = \frac{1-1}{0} = \frac{0}{0} = 1$$

5. (20 分) Prove $\lim_{x\to 2} x^2 = 4$ by definition.

[Q5]

$$\lim_{x\to 2} x^2 = \lim_{x\to 2} x \cdot \lim_{x\to 2} x = 2\cdot 2 = 4$$

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Feb. 23 2026

考試時間: 50 min　　本試卷總分: 100 分　　實得分數

系所／班級: 數學系一　　姓名: 徐庶　　學號: 416100333

Note: No mobile devices. Write inside the box only; no credit for answers outside. L'Hôpital's rule is prohibited.

1. (20 分) 計算以下極極限。

(1) (10 分) $\lim_{x\to 2} 3x^2 - x + 1$.

[Q1-1]

$\lim_{x\to 2} 3x^2 - x + 1 = 3\cdot 2^2 - 2 + 1$

$= 11$

(2) (10 分) $\lim_{x\to 2} \frac{x^2-4}{x-2}$

[Q1-2]

$\lim_{x\to 2} \frac{x^2-4}{x-2} = \lim_{x\to 2} \frac{(x-2)(x+2)}{x-2}$

$= \lim_{x\to 2} x+2 = 4$

2. (20 分) 找出常數 c 的值，使得以下函數在 $x = 3$ 處連續: $f(x) = \begin{cases} cx^2 - 1 & \text{if } x \le 3 \\ 2x + 5 & \text{if } x > 3 \end{cases}$

[Q2]

$f(x)$ is continous at $x=3$ iff $\lim_{x\to 3^+} f(x) = \lim_{x\to 3^-} f(x) = f(3)$

$\therefore c\cdot 9 - 1 = 2\cdot 3 + 5 = 11$

$9c = 12$, $c = \frac{4}{3}$

3. (20 分) 計算以下極限: $\lim_{x\to 0} \frac{\sin(5x)}{x}$

[Q3]

$$\lim_{x\to 0} \frac{\sin 5x}{x} = \lim_{x\to 0} \frac{\sin 5x}{5x} \cdot 5$$

$$\overset{\theta = 5x}{=} \lim_{\theta\to 0} \frac{\sin\theta}{\theta} \cdot 5 = 1 \cdot 5 = 5$$

4. (20 分) 計算以下極限值: $\lim_{x\to 0} \frac{1-\cos(x^2)}{x^3\sin(x)}$

[Q4]

$$\lim_{x\to 0} \frac{1-\cos(x^2)}{x^3\sin(x)} = \lim_{x\to 0} \frac{\frac{1}{2}\sin^2(\frac{x}{2})}{x^3\sin(x)}$$

$$= \lim_{x\to 0} \frac{1}{2} \frac{\sin^2(\frac{x}{2})^2}{(\frac{x}{2})^2} \cdot \frac{(\frac{x}{2})^2}{x^3\sin x} = \lim_{x\to 0} \frac{1}{8} \left(\frac{\sin\frac{x}{2}}{\frac{x}{2}}\right)^2 \cdot \frac{x}{\sin x}$$

$$= \frac{1}{8} \cdot 1^2 \cdot 1 = \frac{1}{8}$$

5. (20 分) Prove $\lim_{x\to 2} x^2 = 4$ by definition.

[Q5]

By definition, $\lim_{x\to 2} x^2 = 4$

iff $\forall \varepsilon > 0, \exists \delta > 0$ s.t

$|x-2| < \delta \Rightarrow |x^2-4| < \varepsilon$.

$|x^2-4| = |x-2||x+2| < \varepsilon$

$\therefore |x-2| < \frac{\varepsilon}{|x+2|}$

Choose $\delta = \frac{\varepsilon}{|x+2|}$

$\therefore |x-2| < \delta$

$\Rightarrow \cancel{|x^2-4| < |x+2| \frac{\varepsilon}{|x+2|}} < \varepsilon$

$|x^2-4| = |x-2||x+2|$

$< \frac{\varepsilon}{|x+2|} \cdot |x+2| = \varepsilon$

Q.E.D.